

Advanced SQL Exercises for Online Retail Store

Exercise 1: Ranking and Window Functions

create database cognizant;

use cognizant;

```
CREATE TABLE Products (  
    ProductID INT PRIMARY KEY,  
    ProductName VARCHAR(50),  
    Category VARCHAR(50),  
    Price DECIMAL(10,2)  
);
```

```
INSERT INTO Products VALUES  
(1, 'Laptop A', 'Electronics', 60000),  
(2, 'Laptop B', 'Electronics', 55000),  
(3, 'Laptop C', 'Electronics', 55000),  
(4, 'Phone A', 'Electronics', 30000),  
(5, 'Shirt A', 'Clothing', 2000),  
(6, 'Shirt B', 'Clothing', 1800),  
(7, 'Shirt C', 'Clothing', 1800),  
(8, 'Shirt D', 'Clothing', 1500),  
(9, 'Book A', 'Books', 500),  
(10, 'Book B', 'Books', 400),  
(11, 'Book C', 'Books', 400);
```

```
SELECT * FROM Products;
```

```
SELECT *
```

```
FROM
```

```
(  
    SELECT ProductID,  
           ProductName,  
           Category,  
           Price,  
           ROW_NUMBER() OVER  
           (  
               PARTITION BY Category  
               ORDER BY Price DESC  
           ) AS RowNum  
    FROM Products  
) AS RankedProducts  
WHERE RowNum <= 3;
```

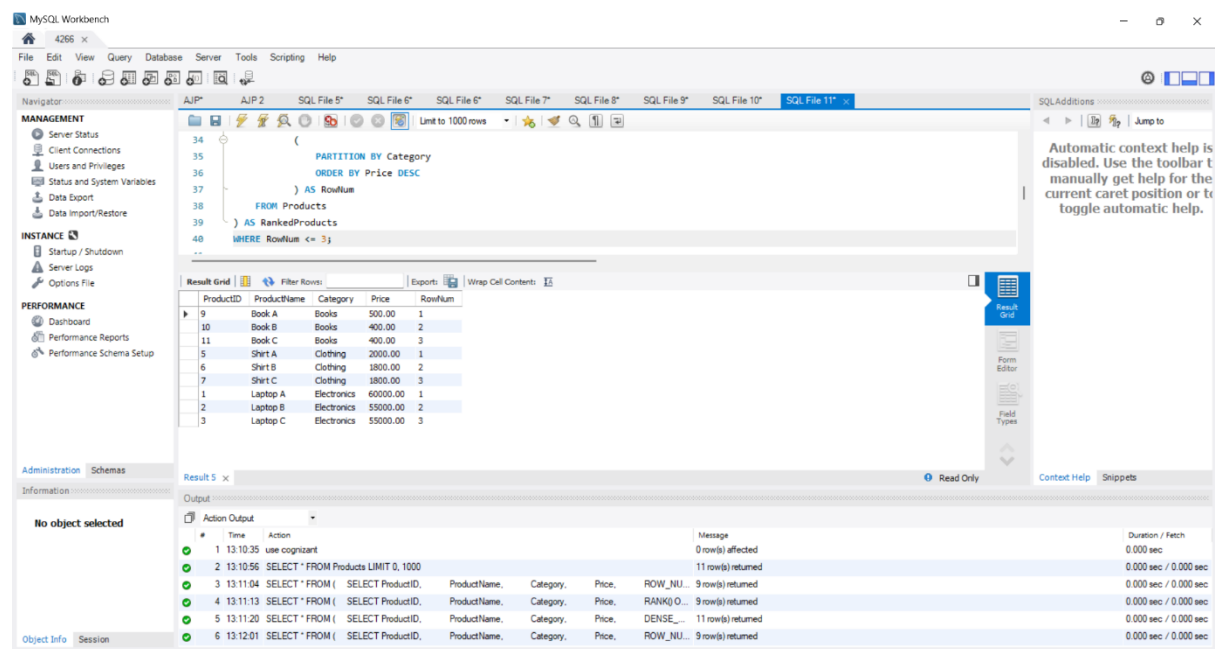
```
SELECT *  
FROM  
(  
    SELECT ProductID,  
           ProductName,  
           Category,  
           Price,  
           RANK() OVER  
           (  
               PARTITION BY Category  
               ORDER BY Price DESC  
           ) AS RankNo  
    FROM Products  
) AS RankedProducts  
WHERE RankNo <= 3;
```

```

SELECT *
FROM
(
    SELECT ProductID,
           ProductName,
           Category,
           Price,
           DENSE_RANK() OVER
           (
               PARTITION BY Category
               ORDER BY Price DESC
           ) AS DenseRankNo
    FROM Products
) AS RankedProducts
WHERE DenseRankNo <= 3;

```

Output:



The screenshot shows the MySQL Workbench interface. The SQL editor contains the following query:

```

34 (
35     PARTITION BY Category
36     ORDER BY Price DESC
37 ) AS RowNum
38 FROM Products
39 ) AS RankedProducts
40 WHERE RowNum <= 3;

```

The Results tab displays the output of the query in a table format:

ProductID	ProductName	Category	Price	RowNum
9	Book A	Books	500.00	1
10	Book B	Books	400.00	2
11	Book C	Books	400.00	3
5	Shirt A	Clothing	2000.00	1
6	Shirt B	Clothing	1800.00	2
7	Shirt C	Clothing	1800.00	3
1	Laptop A	Electronics	60000.00	1
2	Laptop B	Electronics	55000.00	2
3	Laptop C	Electronics	55000.00	3

The bottom pane shows the Output tab with the following log entries:

#	Time	Action	Message	Duration / Fetch
1	13:10:35	use cognizant		0.000 sec
2	13:10:56	SELECT * FROM Products LIMIT 0, 1000	0 row(s) affected	0.000 sec / 0.000 sec
3	13:11:04	SELECT * FROM (SELECT ProductID, ProductName, Category, Price, ROW_NUM...	11 row(s) returned	0.000 sec / 0.000 sec
4	13:11:13	SELECT * FROM (SELECT ProductID, ProductName, Category, Price, RANK() O...	9 row(s) returned	0.000 sec / 0.000 sec
5	13:11:20	SELECT * FROM (SELECT ProductID, ProductName, Category, Price, DENSE...	11 row(s) returned	0.000 sec / 0.000 sec
6	13:12:01	SELECT * FROM (SELECT ProductID, ProductName, Category, Price, ROW_NUM...	9 row(s) returned	0.000 sec / 0.000 sec

MySQL Workbench

4266 x

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
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INSTANCE

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PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Object Info Session

SQL File 11*

```
53 ) AS RankNo
54 FROM Products
55 ) AS RankedProducts
56 WHERE RankNo <= 3;
57
58 SELECT *
59 FROM
```

Result Grid

ProductID	ProductName	Category	Price	RankNo
9	Book A	Books	500.00	1
10	Book B	Books	400.00	2
11	Book C	Books	400.00	2
5	Shirt A	Clothing	2000.00	1
6	Shirt B	Clothing	1800.00	2
7	Shirt C	Clothing	1800.00	2
1	Laptop A	Electronics	60000.00	1
2	Laptop B	Electronics	55000.00	2
3	Laptop C	Electronics	55000.00	2

Result 6 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
2	13.10.56	SELECT * FROM Products LIMIT 0, 1000	11 row(s) returned	0.000 sec / 0.000 sec
3	13.11.04	SELECT * FROM (SELECT ProductID, ProductName, Category, Price, ROW_...	9 row(s) returned	0.000 sec / 0.000 sec
4	13.11.13	SELECT * FROM (SELECT ProductID, ProductName, Category, Price, RANK()...	9 row(s) returned	0.000 sec / 0.000 sec
5	13.11.20	SELECT * FROM (SELECT ProductID, ProductName, Category, Price, DENSE...	11 row(s) returned	0.000 sec / 0.000 sec
6	13.12.01	SELECT * FROM (SELECT ProductID, ProductName, Category, Price, ROW_...	9 row(s) returned	0.000 sec / 0.000 sec
7	13.12.31	SELECT * FROM (SELECT ProductID, ProductName, Category, Price, RANK()...	9 row(s) returned	0.000 sec / 0.000 sec

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.

MySQL Workbench

4266 x

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Information

No object selected

Object Info Session

SQL File 11*

```
66 (
67     PARTITION BY Category
68     ORDER BY Price DESC
69 ) AS DenseRankNo
70 FROM Products
71 ) AS RankedProducts
72 WHERE DenseRankNo <= 3;
```

Result Grid

ProductID	ProductName	Category	Price	DenseRankNo
9	Book A	Books	500.00	1
10	Book B	Books	400.00	2
11	Book C	Books	400.00	2
5	Shirt A	Clothing	2000.00	1
6	Shirt B	Clothing	1800.00	2
7	Shirt C	Clothing	1800.00	2
8	Shirt D	Clothing	1500.00	1
1	Laptop A	Electronics	60000.00	1
2	Laptop B	Electronics	55000.00	2
3	Laptop C	Electronics	55000.00	2
4	Phone A	Electronics	30000.00	3

Result 7 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
3	13.11.04	SELECT * FROM (SELECT ProductID, ProductName, Category, Price, ROW_...	9 row(s) returned	0.000 sec / 0.000 sec
4	13.11.13	SELECT * FROM (SELECT ProductID, ProductName, Category, Price, RANK()...	9 row(s) returned	0.000 sec / 0.000 sec
5	13.11.20	SELECT * FROM (SELECT ProductID, ProductName, Category, Price, DENSE...	11 row(s) returned	0.000 sec / 0.000 sec
6	13.12.01	SELECT * FROM (SELECT ProductID, ProductName, Category, Price, ROW_...	9 row(s) returned	0.000 sec / 0.000 sec
7	13.12.31	SELECT * FROM (SELECT ProductID, ProductName, Category, Price, RANK()...	9 row(s) returned	0.000 sec / 0.000 sec
8	13.12.58	SELECT * FROM (SELECT ProductID, ProductName, Category, Price, DENSE...	11 row(s) returned	0.000 sec / 0.000 sec

SQLAdditions

Automatic context help is disabled. Use the toolbar to manually get help for the current caret position or to toggle automatic help.